

17

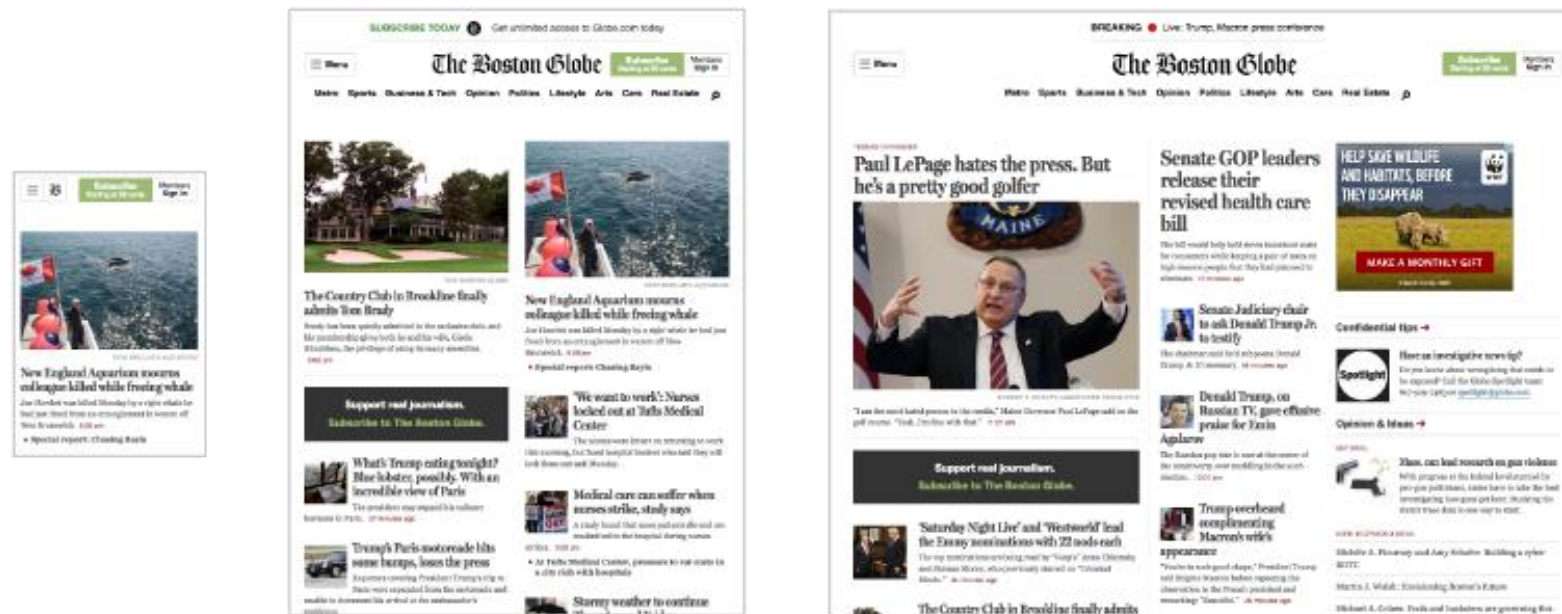
RESPONSIVE WEB DESIGN

Introduction to RWD

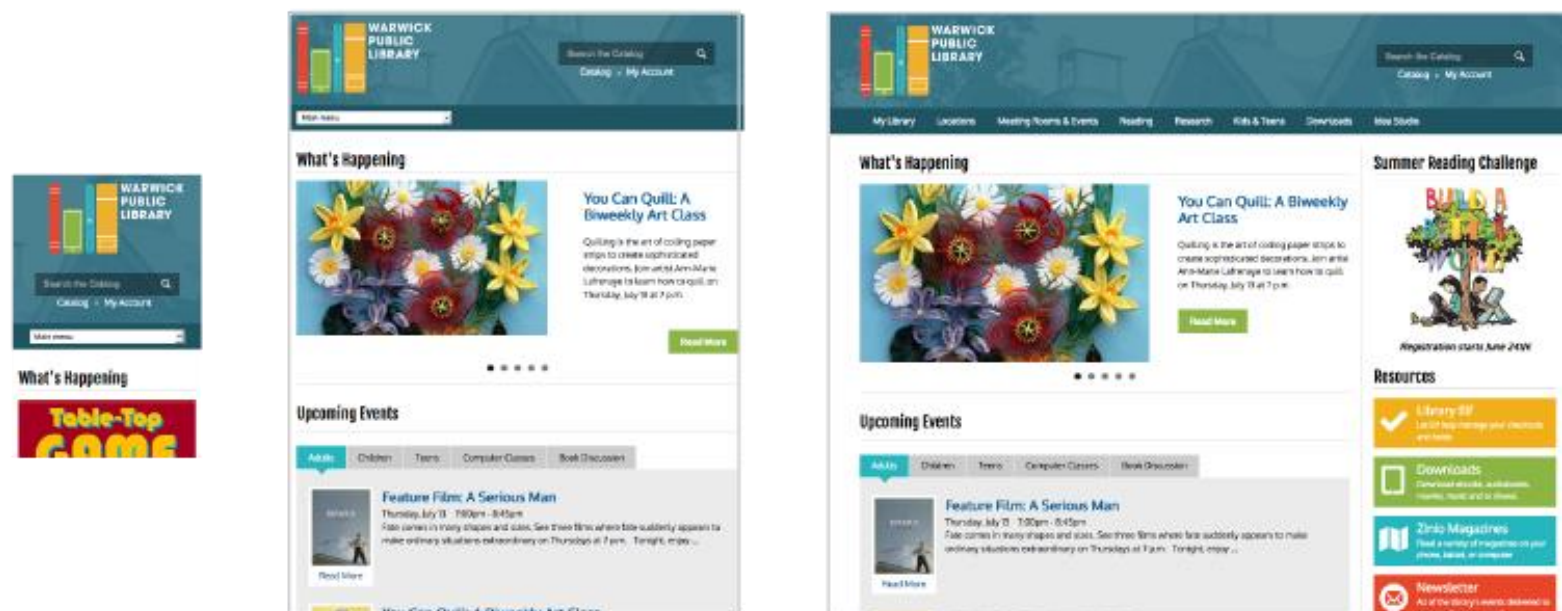
- RWD is a design and production approach that allows a web page to adapt to the screen it is viewed on
- The same HTML source is styled differently based on the size of the screen
- It ensures mobile users get the same content as other users, with a presentation that optimizes usability

Introduction to RWD (cont'd)

The Boston Globe (*bostonglobe.com*)



Warwick Public Library (*warwicklibrary.org*)



The Components of RWD

Ethan Marcotte defines the three core components as follows:

- **A flexible grid**

Elements are sized so they squeeze and flow into the available browser space.

- **Flexible images**

Images and other embedded media scale to fit their containing elements.

- **CSS media queries**

These are a way to test for browser features such as viewport width and deliver styles accordingly.

Setting the Viewport

- Mobile browsers render pages on a canvas called the **viewport** and then shrink it to fit the screen (**device width**) – essentially done this way to ensure that mobile browser users are able to comfortably view websites that aren't responsive
- For example, a web page be rendered on a 980-pixel viewport and shrunk down to fit a 480-pixel wide screen.
- **To make the viewport the same size as the device width:**

```
<meta name="viewport"  
      content="width=device-width, initial-scale="1">
```

[https://developer.mozilla.org/en-US/docs/Web/HTML/Viewport meta tag](https://developer.mozilla.org/en-US/docs/Web/HTML/Viewport_meta_tag)

Setting the Viewport (cont'd)

By default, the viewport shrinks to the size of the screen.



With the viewport meta tag, the viewport is created at the same size as the screen.

```
<meta name="viewport"
      content="width=device-width,
              initial-scale=1">
```



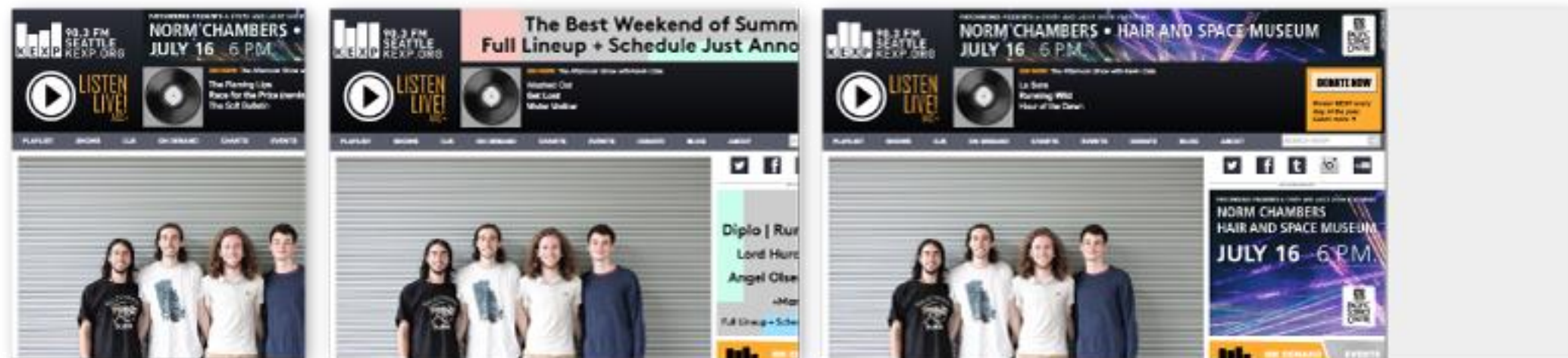
Fluid Layout

Fluid layouts fill the viewport proportionally.



w3c.org

Fixed layouts stay the same size and may get cut off or leave extra space.



kexp.org

Fluid Layout

- **Fluid layouts** are the best approach for designing for a wide range of screen widths
- For flexibility, the easiest way is to use percentage measurements
 - Can be applied to all layout methods (float, position, flex and grid)

Making Images Flexible

To make images scale down to fit the size of their container:

```
img { max-width: 100%; }
```

In this book, Philadelphia Ice Creams, comprising the first group, are very palatable, but expensive. In many parts of the country it is quite difficult to get good cream. For that reason, I have given a group of creams, using part milk and part cream, but it must be remembered that it takes smart "juggling" to make ice cream from milk. By far better use condensed milk, with enough water or milk to rinse out the cans.

Ordinary fruit creams may be made with condensed milk at a cost of about fifteen cents a quart, which, of course, is cheaper than ordinary milk and cream.



If pure raw cream is stirred rapidly, it swells and becomes frothy, like the beaten whites of eggs, and is "whipped cream." To prevent this in making Philadelphia Ice Cream, one-half the cream is scalded, and when it is very cold, the remaining half of raw cream is added. This gives the smooth, light and rich consistency which makes these creams so different from others.

The time for freezing varies according to the quality of cream or milk or water; water ices require a longer time than ice creams. It is not well to freeze the mixtures too rapidly; they are apt to be coarse, not smooth, and if they are churned before the mixture is icy cold they will be greasy or

Philadelphia

In this book, Philadelphia Ice Creams, comprising the first group, are very palatable, but expensive. In many parts of the country it is quite difficult to get good cream. For that reason, I have given a group of creams, using part milk and part cream, but it must be remembered that it takes smart "juggling" to make ice cream from milk. By far better use condensed milk, with enough water or milk to rinse out the cans.

Ordinary fruit creams may be made with condensed milk at a cost of about fifteen cents a quart, which, of course, is cheaper than ordinary milk and cream.



If pure raw cream is stirred rapidly, it swells and becomes frothy, like the beaten whites of eggs, and is "whipped cream." To prevent this in making Philadelphia Ice Cream, one-half the cream is scalded, and when it is very cold, the remaining half of raw cream is added. This gives the smooth, light and rich consistency which makes these creams so different from others.

In this book, Philadelphia Ice Creams, comprising the first group, are very palatable, but expensive. In many parts of the country it is quite difficult to get good cream. For that reason, I have given a group of creams, using part milk and part cream, but it must be remembered that it takes smart "juggling" to make ice cream from milk. By far better use condensed milk, with enough water or milk to rinse out the cans.

Ordinary fruit creams may be made with condensed milk at a cost of about fifteen cents a quart, which, of course, is cheaper than ordinary milk and cream.



If pure raw cream is stirred rapidly, it swells and becomes frothy, like the beaten whites of eggs, and is "whipped cream." To prevent this in making Philadelphia Ice Cream, one-half the cream is scalded, and when it is very cold, the remaining half of raw cream is added. This

Media Queries

A **media query** is a special rule that tests for certain conditions before applying the styles it contains:

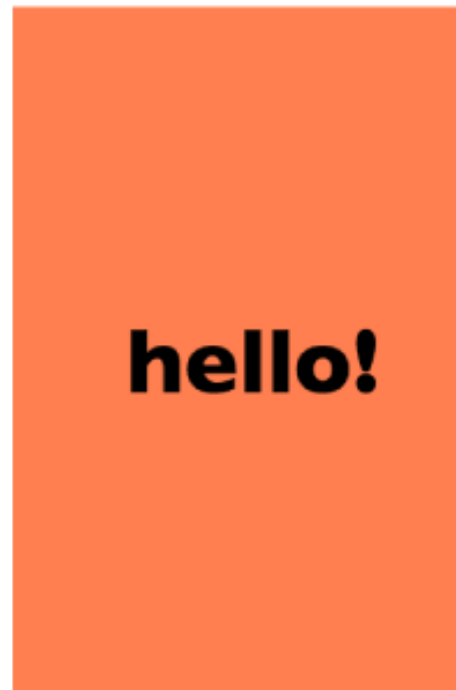
```
@media type and (feature: value) {  
    /* styles for browsers that meet this criteria */  
}
```

A few query examples:

- Is it on a screen or printed?
- Is the browser at least a certain width?
- Is the orientation landscape or portrait?

Media Query Example

```
@media screen and (orientation: landscape) {  
  body {  
    background: skyblue;  
  }  
}  
@media screen and (orientation: portrait) {  
  body {  
    background: coral;  
  }  
}
```



When the viewport is in portrait mode, the background color is "coral."



When the viewport is in landscape mode, the background color is "skyblue."

Media Types

- The first part of a media query identifies the **media type**.
- The current media types are **all**, **print**, **screen**, and **speech**.

Example:

```
@media print {  
    /* print-specific styles between the brackets */  
}
```

- Media types were introduced in CSS2.

Media Feature Queries

CSS3 introduced media **feature queries** that test for a particular feature of a viewport or device.

Example:

h1 headings display in Lobster font only when the viewport is *at least 40em wide*.

```
h1 {  
  font-family: Georgia, serif;  
}  
@media screen and (min-width: 40em) {  
  h1 {  
    font-family: 'Lobster', cursive;  
  }  
}
```

NOTE: `min-width` and `max-width` queries are most useful for RWD.

Using Media Queries

In a style sheet:

- Put an `@media` rule in a style sheet to deliver styles based on viewport width.
- Make sure the `@media` rule comes after your baseline styles.

With external style sheets:

- Link or `@import` separate style sheets based on a query:

```
<link rel="stylesheet" href="styles.css">  
<link rel="stylesheet" href="2column-styles.css"  
      media="screen and (min-width:1024px)">
```


Breakpoints

- A **breakpoint** is the point at which we use a media query to introduce a style change
- Best practice is to create breakpoints for individual parts of the page rather than the entire page at once
- Designers generally create the narrow-screen version first and then change components as needed as the screen grows larger
- Em-based breakpoints introduce layout changes proportional to font size

Designing Responsively

You need to pay attention to the following aspects of the page in a responsive design:

- Content hierarchy
- Layout
- Typography
- Navigation
- Images
- Special content like tables and forms

Content Hierarchy

- The biggest challenge is deciding **what is visible in the first screen** on small devices.
- You should strive for **content parity** across devices. Everyone should have access to the same information, even if it's not in exactly the same place.
- Content organization is the most important part of site planning. It's often handled by information architects or content strategists.

Layout

- RWD (responsive web design) is based on **fluid layouts**.
- The most common approach is to **start with one column** on small devices and **add columns as space is available**

Typography

- Typography should be optimized for different screen sizes.
- Avoid complicated font styles on small screens.
- Line height should generally be larger on larger screens.
- Left and right margins can increase on larger screens.

Site Testing

Testing sites in a wide variety of browsing environments is critical:

- **Real devices**

The best way to test is on actual smartphones, tablets, etc.

- **Emulators**

Use a desktop application that simulates device hardware and software.

- **Browser testing services**

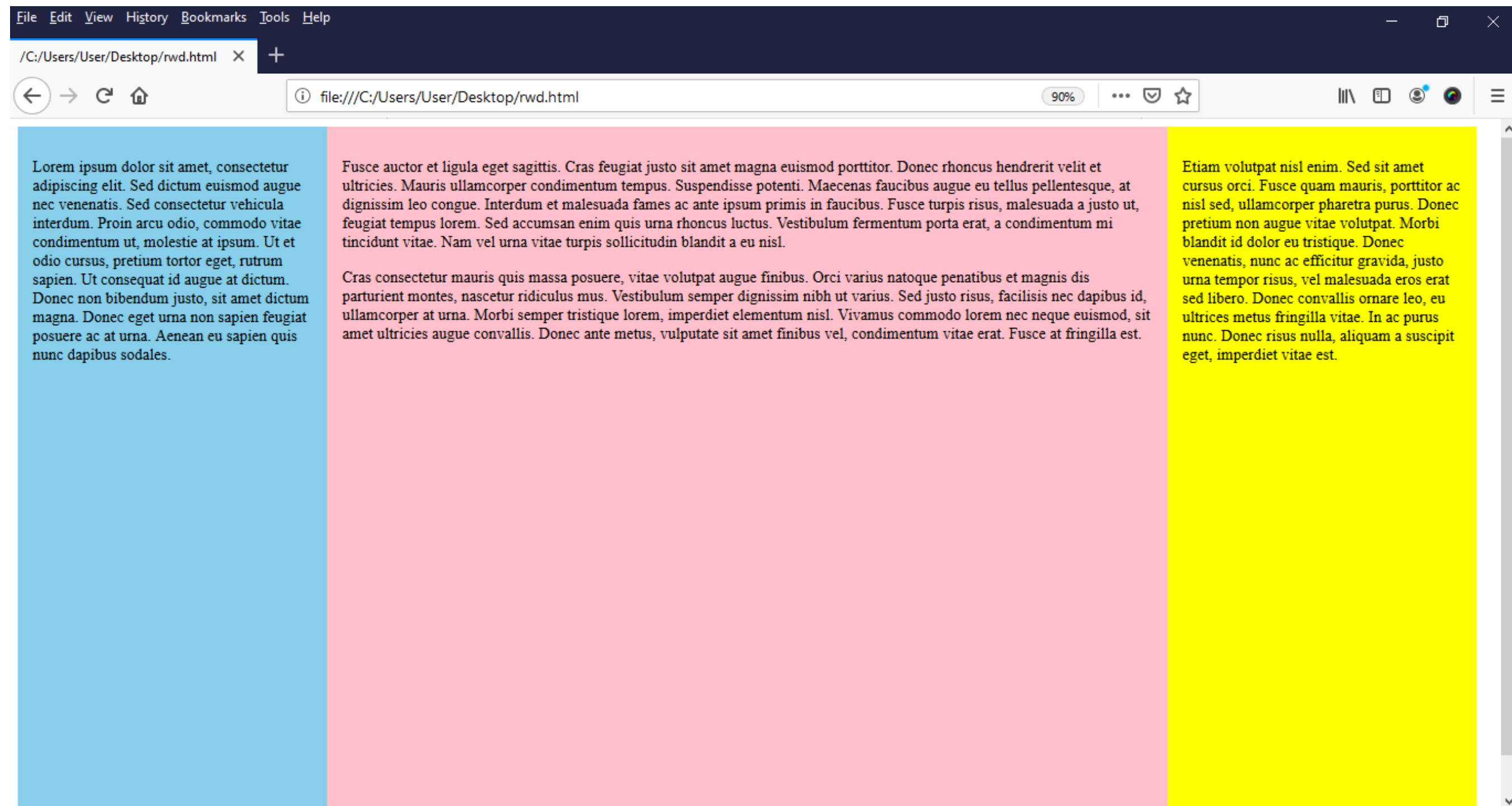
Subscription-based services show how your site looks on a huge variety of browsers.

- Example

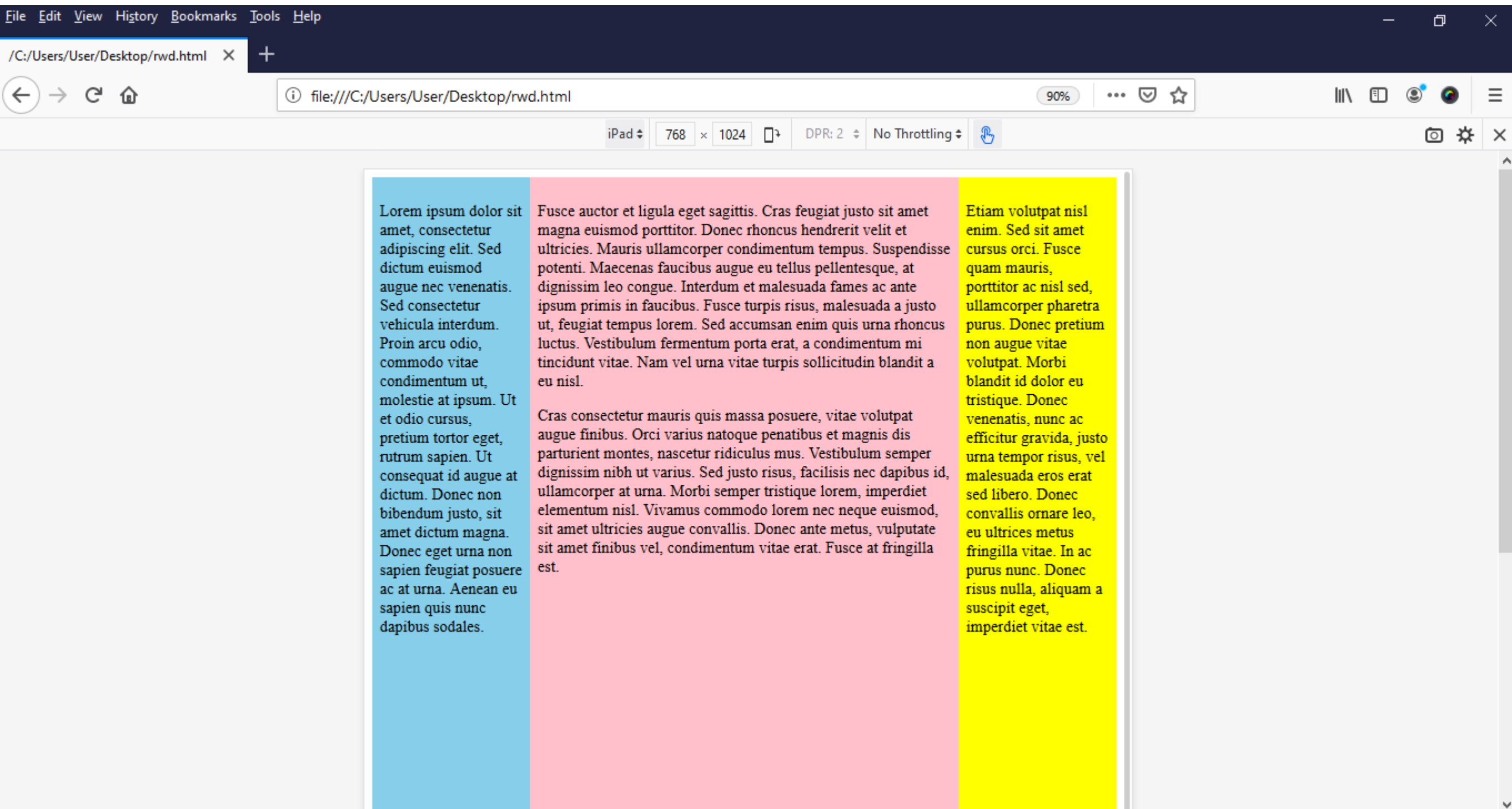
Responsive Web Design

- Create a layout which is three column on laptop/desktop/iPad sized screens but changes to stacked sections in on iPhone size
- Generate your own text to use in the divs
- Test using the responsive web design mode in Firefox
- Sample screenshots ahead

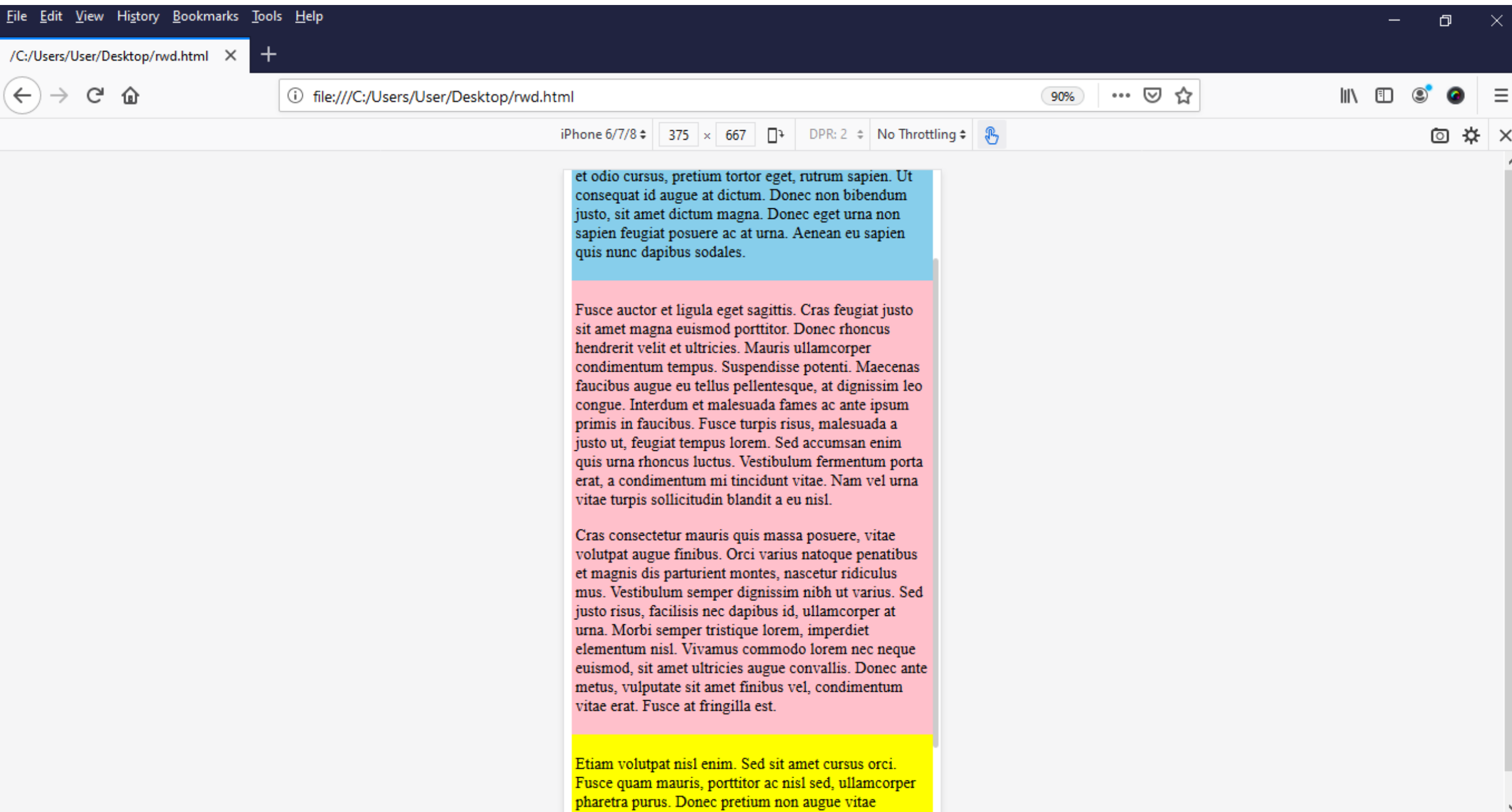
Laptop/Desktop



iPad



iPhone



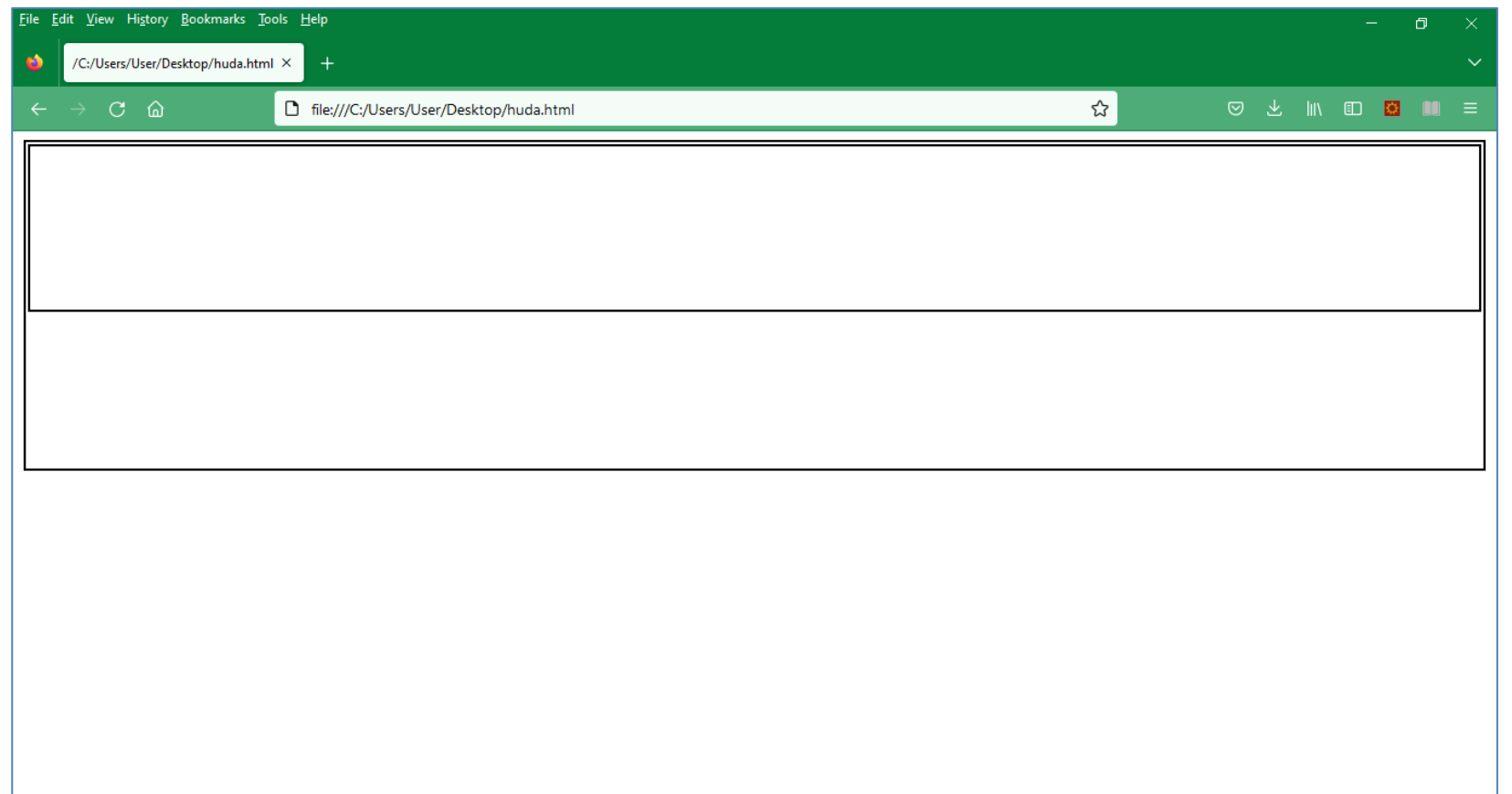
Complete and test on browsers

```
<style>
  div{ padding: 1%; }
  #divA { background-color: skyblue; }
  #divB{ background-color: pink; }
  #divC{ background-color: yellow;}

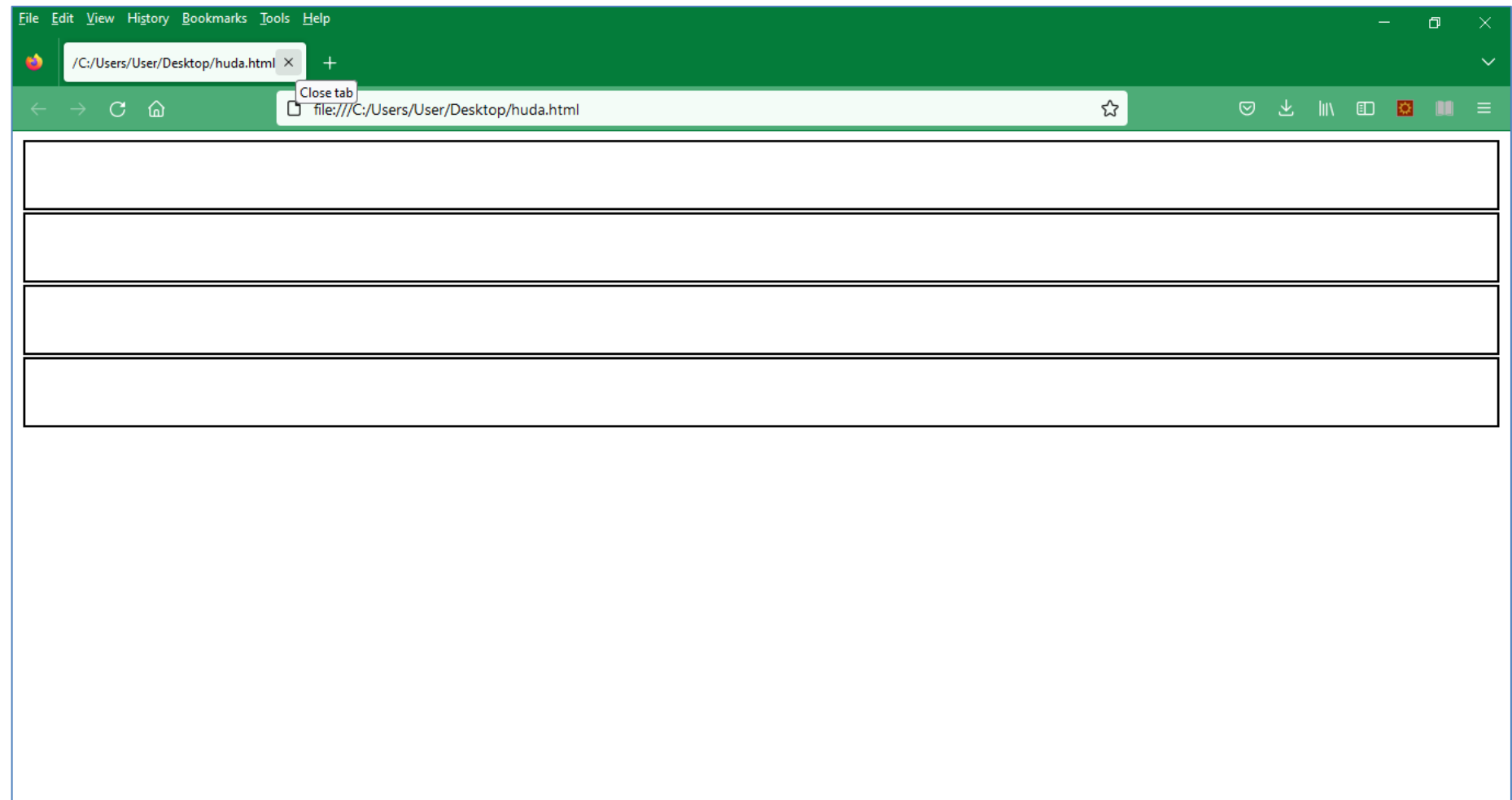
  @media screen and (min-width: 650px) {
    div{ height: 100vh; }
    #divA{
      width: 19%;
      float: left;
    }
    #divB{
      width: 55%;
      float: left;
    }
    #divC{
      width: 19%;
      float: left;
    }
  }
</style>
```

EXAMPLES

```
<html>
<head>
  <style>
    div {height: 50%;
        border: 2px solid black;
        margin: 2px;}
  </style>
</head>
<body>
  <div>
    <div></div>
  </div>
</body>
</html>
```




```
<html>
<head>
  <style>
    div {height: 10%;
        border: 2px solid
black;
        margin: 2px;}
  </style>
</head>
<body>
  <div></div>
  <div></div>
  <div></div>
  <div></div>
</body>
</html>
```




```
<html>
<head>
  <style>
    div {height: 20%; width:20%;
        border: 2px solid
black;

        position: absolute;
        margin: 3px;

    }
    #a {top: 0px; left: 0px;}
    #b {top: 0px; right: 0px;}
    #c {bottom: 0px; right:
0px;}
    #d {bottom: 0px; left: 0px;}
  </style>
</head>
<body>
  <div id="a">A</div><div
id="b">B</div>
  <div id="c">C</div><div
id="d">D</div>
</body>
</html>
```

